

Tallying

Results from a survey can be organized into a chart. The left-hand column shows the groups of data from the questionnaire. A simple way to record the results is by making a tally mark in the chart for each answer. To tally, mark a line for each unit and cross through the lines when 5 is reached.

making tally marks in groups of five makes chart easier to read; the line that goes across is the 5th

Soft drink	Tally
Cola	I
Orange juice	I
Apple juice	
Pineapple juice	I
Milk	
Other	I

△ Tally chart

This tally chart shows the results of the survey with tally marks.

Soft drink	Tally	Frequency
Cola	I	6
Orange juice	I	11
Apple juice		2
Pineapple juice	I	1
Milk		2
Other	I	1

△ Frequency table

Counting the tally marks for each group, the results (frequency) can be entered in a separate column to make a frequency table.

Tables

Tables showing the frequency of results for each group are a useful way of presenting data. Values from the frequency column can be analyzed and used to make charts or graphs of the data. Frequency tables can have more columns to show more detailed information.

Drink	Frequency
Cola	6
Orange juice	11
Apple juice	2
Pineapple juice	1
Milk	2
Other	1

△ Frequency table

Data can be presented in a table. In this example, the number of children that chose each type of drink is shown.

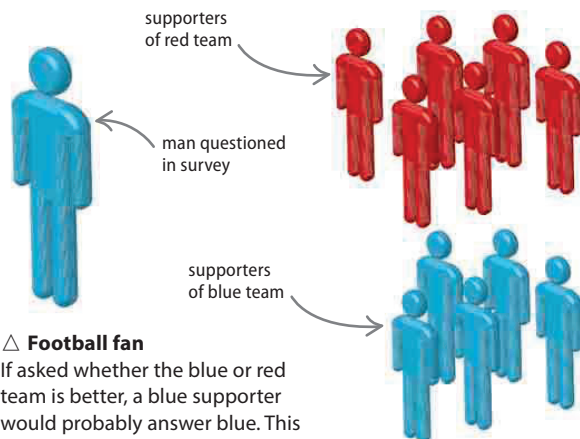
Drink	Boy	Girl	Total
Cola	4	2	6
Orange juice	5	6	11
Apple juice	0	2	2
Pineapple juice	1	0	1
Milk	1	1	2
Other	1	0	1

△ Two-way table

This table has extra columns that break down the information further. It also shows the numbers of boys and girls and their preferences.

Bias

In surveys it is important to question a wide selection of people, so that the answers provide an accurate picture. If the survey is too narrow, it may be unrepresentative and show a bias toward a particular answer.



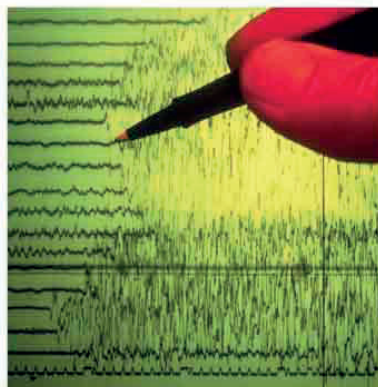
△ Football fan

If asked whether the blue or red team is better, a blue supporter would probably answer blue. This would be regardless of whether the reds had proved their superiority.

LOOKING CLOSER

Data logging

A lot of data is recorded by machines—information about the weather, traffic, or internet usage for instance. The data can then be organized and presented in charts, tables, and graphs that make it easier to understand and analyze.



◁ Seismometer

A seismometer records movements of the ground that are associated with earthquakes. The collected data is analyzed to find patterns that may predict future earthquakes.